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Quantum-sensing-enabled in situ imaging of free radicals in chemical reactions

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In situ observation of chemical reactions with high sensitivity and high spatiotemporal resolution is a long-standing goal in chemistry. Up to now, only a few, very specific systems can be studied. High-resolution operando imaging of chemical reactions typically requires a fluorescent probe to be included in the reaction. Here we introduce an operando microscopy methodology, namely quantum-sensing-enabled chemical operando microscopy (QCOM), in which reaction-induced local physical field changes are intrinsically transformed into dynamic imaging contrast using nitrogen vacancy centres in diamond. QCOM simultaneously satisfies the criteria of detection sensitivity (around four free radicals per pixel), spatial resolution (full-width-at-half-maximum resolution of around 312 nm) and temporal resolution (~10–240 ms). This has enabled the direct, quantitative, operando imaging of free-radical generation in the photocatalytic hydrolysis of TiO_2, revealing an unexpected spatiotemporal sequential activation effect. Our work combines quantum sensing with chemical reaction imaging, offering a comprehensive in situ methodology to explore spatiotemporal phenomena in chemistry.

Unlocking undetected physical and chemical transformations during chemical reactions is vital to identify catalytic mechanisms and design efficient chemistry. As a representative case, free radicals, which are short-lifetime molecular species with at least one unpaired electron, often serve as reactive intermediates in chemical reactions. Therefore, probing free radicals is of fundamental importance for elucidating reaction sites, mechanisms, dynamics and functions in both chemistry and biology[1]. Electron spin resonance (ESR) spectroscopy provides a direct approach to explicitly identify and quantify free radicals in chemical reactions[2], but it requires large ensembles of radicals to obtain ensemble average information and lacks spatial resolution. In contrast, microscopy can provide essential spatial information with respect to reaction sites, which is crucial to build structure–activity relationships at various levels of organization[3–9]. Nevertheless, conventional optical imaging of chemical reactions requires the use of luminescent probes[4,5,10,11] that must be chemically coupled to the reaction itself, limiting its use as the majority of chemical reactions do not exhibit luminescence. As such, the dynamic imaging of classical free-radical reactions remains an enormous challenge.

Chemical transformations are typically associated with a change in a physical quantity, such as a variation in temperature during endothermic or exothermic reactions, current flowing during electrochemical reactions and a fluctuation of electromagnetic noise during free-radical reactions. Here we propose an operando microscopy methodology that characterizes chemical reactions by in situ sensing of these localized physical changes using atomic-scale quantum sensors and directly transforms this information into image contrast with both high spatial and temporal resolution. We tested this quantum-sensing-enabled

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reaction imaging concept on a classical free-radical reaction system, which enabled the direct, operando, spatiotemporally resolved imaging of free radicals in a photocatalytic reaction and allowed us to intrinsically study reaction sites, quantities, activation dynamics and free-radical species. These observations further uncovered a spatiotemporal sequential activation in TiO_2 photocatalysis.

Due to its long spin coherence time and chemical inertness, the atomic-scale nitrogen vacancy (NV) centre has enabled high-sensitivity quantum sensing to measure multiphysical quantities and spin interactions down to the single spin level under ambient conditions[12,13] and achieved static imaging of paramagnetic ions[14,15] and dynamic detection of free radicals at single sites without spatial resolution[16,17]. However, developing a spatiotemporal free-radical reaction imaging system with NV centres necessitates a quantum sensing–chemical reaction interface with in situ capabilities. To meet this need, several experimental challenges must be addressed, such as the coupling of the chemical reaction system to widefield quantum sensing while simultaneously satisfying spatial resolution, temporal resolution and detection sensitivity. We pursued this task by implementing the above-mentioned microscopy concept in a custom-built set-up. By probing the coupling between NVs and radical spins, we conducted operando longitude relaxation (T1) imaging, demonstrating the dynamic monitoring of free-radical generation during the photocatalytic hydrolysis of TiO_2 and uncovering a spatiotemporal sequential activation effect through in situ observations.

Results

Probing chemical reactions with quantum sensors

We developed quantum-sensing-enabled chemical operando microscopy (QCOM) to spatiotemporally resolve free radicals in chemical reactions using NV centres, facilitated by spin manipulation, parallel optical readout and dynamic observation. Our concept is illustrated in Fig. 1a–c and an image of the custom-designed instrument is shown in Supplementary Fig. 1, which integrates a multichannel synchronous control system, a widefield optical microscope, a microwave system, a magnet (Supplementary Fig. 2) and an in situ chemical reaction chamber equipped with real-time temperature, fluidic and light control (Supplementary Fig. 3). We created shallow NV centres in diamond using ion implantation, resulting in spatially uniformly distributed and temporally stable (Supplementary Fig. 4–6) NV samples for widefield ($100 \times 100\ \mu m^2$) free-radical imaging (Methods and Supplementary Note 1). Next, we probed the reactions occurring at the chemical interface using NV-based electron spin sensors in which a 532-nm laser was used for the initialization and readout of spin states, while a microwave was fed into the system through a microfabricated waveguide (Supplementary Note 2), allowing precise modulation of the spin state of the NV centre. As shown in Fig. 1b, each NV centre hosts a spin-1 electronic ground state with three energy levels, which can be tuned by interactions with multiphysical fields such as temperature[18], stress[19], electric field[20], magnetic field[21–24] and external spins[25,26], reflected in a change in the intensity of its fluorescence signal.

We employed a classical photocatalytic free-radical reaction in our experiments: a TiO_2 photocatalyst, activated by UV light ($\lambda = 365$ nm, Supplementary Fig. 7), generates an electron–hole pair, which reacts with water and oxygen adsorbed on the catalyst and produces the corresponding hydroxyl (·OH) and superoxide ($O_2^{\cdot-}$) radicals[27] (Supplementary Note 3).

To probe the free radicals, as the target free radical has an electron spin that generates an effective magnetic field[14], we could use this interaction to sense the external paramagnetic noise perturbations at the NV centre, as defined by a coupled Hamiltonian (H) describing the free-radical–NV centre electron spin system (equation (1)):

$$H = \Omega S_z^2/\hbar - \gamma_{NV} S_z B^{bias} - \gamma_{NV} \mathbf{S} \cdot \mathbf{B}(t) \quad (1)$$

where $\Omega = 2\pi \times 2.87$ GHz is the zero-field splitting of the NV centre, with the NV centre axis defined in the z direction, S_z is the z component of $\mathbf{S}$, which is the spin of the NV centre, $\gamma_{NV} = -2\pi \times 28\ \text{GHz T}^{-1}$ is the gyromagnetic ratio of the spin, B^{bias} is the external magnetic field aligned with the z axis, $\mathbf{B}(t)$ is the total time-dependent magnetic field generated by free radicals and $\hbar = h/2\pi$, where h is Planck's constant.

To probe this interaction, we measured the T1 relaxation time. In principle, T1 is the spin–lattice relaxation time of a NV's electron spin, which is highly sensitive to magnetic fluctuations from nearby environments, such as electron spins[28,29]. When a transient free radical approaches the NV centre, its unpaired electron generates magnetic noise that accelerates the NV's spin relaxation via electron–electron coupling, leading to changes in the T1 relaxation time[14,15] (Supplementary Note 4 and Supplementary Fig. 8). Imaging these processes under operando conditions requires not only sensing of the free-radical–NV interactions but also high spatial and temporal resolution. Accordingly, we developed a fast relaxation imaging (FRI) sequence (Fig. 1d, Supplementary Note 4 and Supplementary Fig. 9), which we used in combination with a scientific complementary metal oxide semiconductor (sCMOS) camera with a high quantum yield (>95%) and imaging speed (200 frames per second) to achieve high spatiotemporal resolution imaging of free radicals. Our FRI sequence involves a trade-off between sampling points and image contrast and uses a synchronization control system to minimize the aberrations introduced during data acquisition, leading to the contrast of the T1 relaxation curves at each image pixel, as shown in Fig. 1d. When switching on the 365-nm light to trigger the photocatalytic reaction[27], a significant change in T1 can be seen on a catalyst particle (Fig. 1e,f). We designed a series of control experiments that indicated that the changes in the T1 contrast are caused by the generation of free radicals rather than photothermal effects or changes in the refractive index (Supplementary Fig. 10). Moreover, we evaluated the spatial resolution of QCOM, featuring a representative full-width-at-half-maximum (FWHM) resolution of ~312 nm and a double-peak resolvable separation of ~335 nm (Supplementary Fig. 11 and Supplementary Note 5).

Operando quantitative imaging of free-radical generation

We next demonstrated the direct imaging of free radicals in the photocatalytic reactions on both single TiO_2 nanowires (Fig. 2a) and nanoparticles (Fig. 2b). In addition, the power density of the UV illumination could be used to effectively control the rate of electron–hole pair separation on a photocatalyst, thereby affecting the reaction kinetics. This was visualized by monitoring the generation of free radicals on individual catalyst particles (Fig. 2c–e). An increase in light power led to an increase in free-radical production (Fig. 2e). To demonstrate the quantitative capability of QCOM, we employed a fluorescent probe (2-hydroxyterephthalic acid (HTA)) to quantify the concentration of free radicals (Fig. 2d, Methods and Supplementary Fig. 12). First, we established the relationship between the fluorescence intensity and HTA concentration, obtaining a standard calibration curve. As each radical corresponds to one molecule of HTA upon capture, this relationship was used to calibrate the radical concentration generated in the same chemical reaction system monitored by QCOM. Consequently, the correlation between radical concentration and relaxation time was obtained (Fig. 2d). We further derived a quantitative model describing this relationship, as detailed in equation 2) (see Supplementary Note 6 and Supplementary Figs. 13 and 14 for details):

$$\frac{1}{T1} = \frac{1}{T1^{base}} + 3\gamma_{NV}^2 B_\perp^2 \frac{\tau_c}{1 + \Omega^2 \tau_c^2}, \quad (2)$$

where $T1^{base}$ is the T1 of the NV sensor without perturbation of free radicals, $B_\perp^2$ is the variance of the transverse magnetic field at the NV site and τ_c is the correlation time of the free-radical ensemble. The limit of detection (LOD) of our method can then be derived from this

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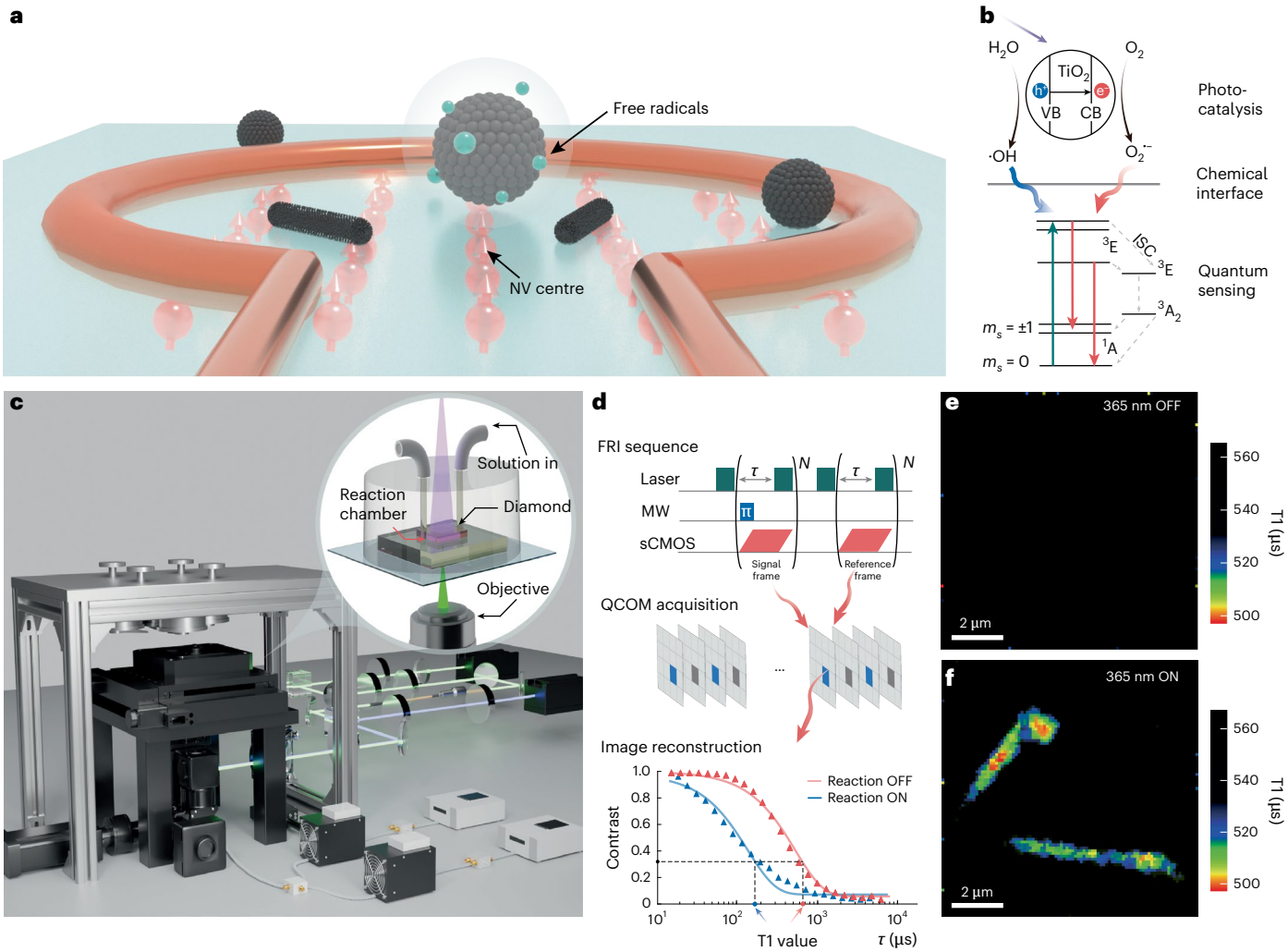

Fig. 1 | Quantum sensing of free radicals in photocatalytic reactions. **a**, Schematic of the NV quantum sensing–photocatalytic reaction interface. **b**, Generation of free radicals in the photocatalysis of TiO_2 and electron energy level diagram of the NV centre. MW, microwave; VB, valence band; CB, conductance band; H^+, hole; e^-, electron; ISC, intersystem crossing; m_s, spin magnetic quantum number; the sublevels are labeled A and E, according to their symmetry representations. **c**, Instrumental design of the QCOM set-up. Inset: schematic of the in situ chemical reaction chamber. **d**, Workflow of free-radical imaging via QCOM, including the FRI sequence, image acquisition and image reconstruction of free-radical distributions by subtracting the T1 value from the acquired curves. *N*, number of loops in the sequence; *τ*, length of dark pulse. **e**,**f**, Free-radical images acquired when the 365-nm light-emitting diode (LED) is switched off (**e**) and on (**f**).

quantitative relationship, yielding a value of about four radical species per pixel (Supplementary Fig. 15 and Supplementary Note 7). It is worth noting that the depth and density of NVs in diamond samples can also affect the quantitative relationship and the LOD (Supplementary Figs. 16 and 17 and Supplementary Note 8). Moreover, owing to the operando observations, we can directly evaluate the effect of other physiochemical factors that evolve in this reaction, such as the pH effect. Figure 2f shows the T1 values measured at different pH, indicating that an alkaline chemical environment providing more hydroxide ions could accelerate the generation of free radicals[30].

Spatiotemporally resolved catalytic activation

Our approach simultaneously allows temporal resolution to reveal reaction dynamics, which is achieved by adopting a 'moving window average' method along with image denoising (Methods, Supplementary Fig. 18, Supplementary Table 1 and Supplementary Note 5). We carried out temporally resolved operando imaging on single catalyst particles, as shown in Fig. 3a. Reversible switching of the UV light actively drives the photocatalytic reactions into on and off states and our dynamic observation effectively captures this swift response (Supplementary Fig. 19).

Next, to provide spatiotemporal insight into the long-term operando catalytic operation dynamics, the catalytic reactions were monitored under continuous UV illumination for 20 min (Fig. 3a). In the classical photocatalysis mechanism, upon UV excitation, the TiO_2 catalyst should be fully activated immediately after the ultrafast photoinduced electron–hole separation and reaction dynamics (picosecond to millisecond timescale)[31,32]. Unexpectedly, we observed rather slow activation dynamics (10^2 s scale) for all the TiO_2 catalysts that we imaged, in which the total area of free-radical generation hotspots increased with time along a single TiO_2 nanowire. More interestingly, by segmenting a single nanowire into four intraparticle regions, we could analyse the temporal evolution of T1 in the four regions, which revealed spatially heterogeneous reaction kinetics within individual catalysts (Fig. 3b–d). In general, the overall upward tendency of the curves indicates a decrease in T1 contrast, which corresponds to the increasing generation of radicals during the reaction process. Some fluctuations may relate to the consumption of free radicals in solution or the adsorption–desorption of reactants on the catalysts[33,34]. Moreover, Pearson cross-correlation analysis was performed to evaluate the correlation between time and space in the generation of free radicals

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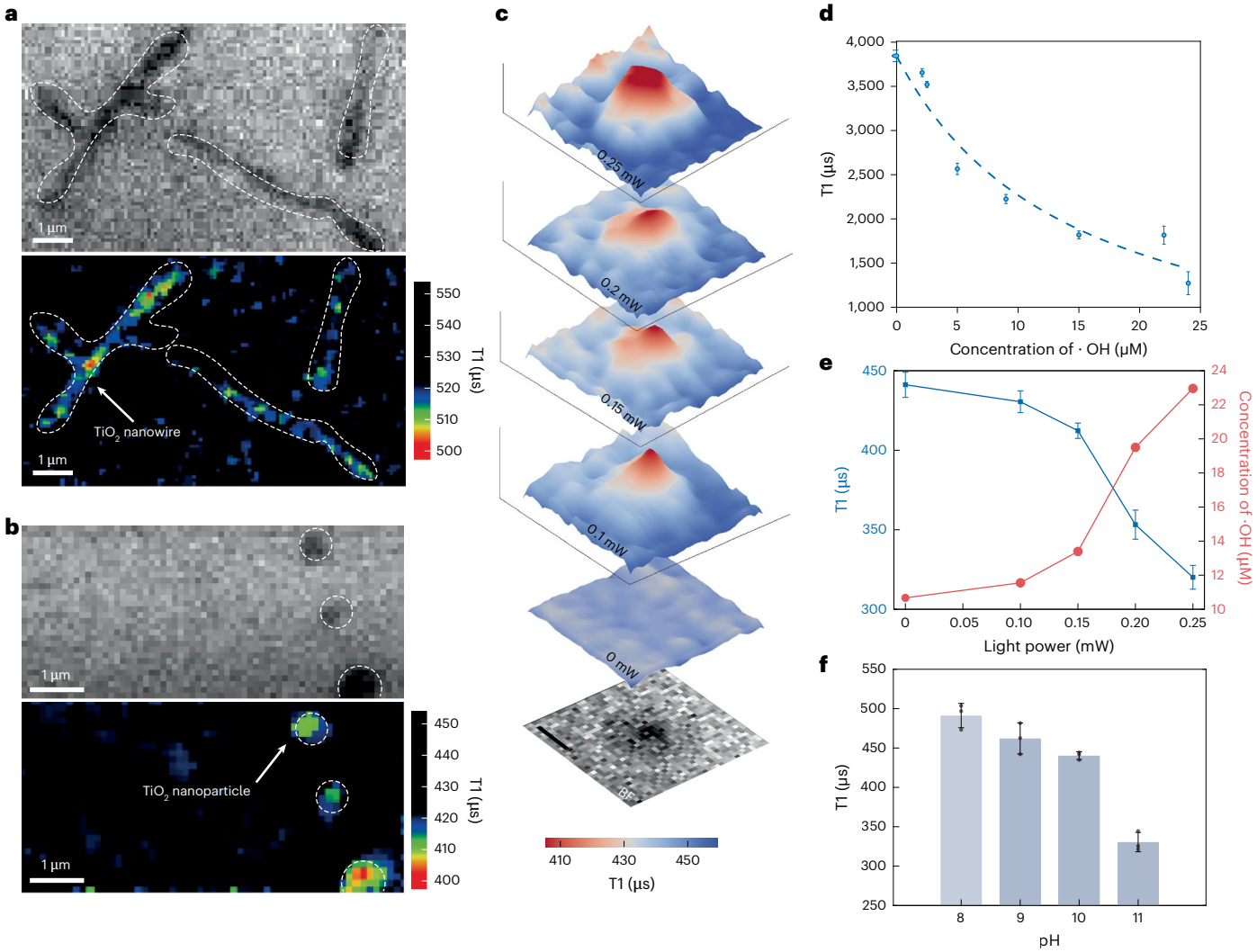

Fig. 2 | Quantitative imaging of free radicals in the photocatalytic hydrolysis of TiO_2. **a**,**b**, Bright-field and free-radical images of the photocatalytic reactions on TiO_2 nanowires (**a**) and TiO_2 nanoparticles (**b**). **c**, Free-radical imaging of the photocatalytic reaction under varying 365-nm LED powers from 0 to 0.25 mW and the bright field image (BF) of same region. Scale bar, 1 μm. **d**, Relationship between T1 values and the hydroxyl radical concentration. **e**, Variation in T1 values (blue) and hydroxyl radical concentration (red) in photocatalytic reactions conducted with increasing light power. **f**, Change in T1 values in the photocatalytic reactions with solution pHs ranging from 8 to 11. In **d**–**f**, all data are presented as the mean ± standard deviation from three independent experiments ($n = 3$).

(Supplementary Fig. 20). Surprisingly, the time correlation curves for the four individual intraparticle regions in Fig. 3e–g reveal a sequential activation phenomenon (Supplementary Fig. 21). This correlated effect is further captured in the exponential decay of the intraparticle space correlation analysis, revealing a characteristic distance of ~295 nm with correlation coefficients in the range of 0.70–0.92 (Fig. 3h).

This implies that the active sites within a single TiO_2 nanowire are sequentially activated by earlier-activated neighbouring sites, providing an insight into the classical picture of photocatalytic free-radical reaction dynamics. This sequential activation phenomenon occurring on the photocatalyst is different from the known catalytic migration phenomena observed on metal substrates, such as the hydrogen spillover effect[35] and the catalyst communication effect[36]. We also conducted photocurrent measurements, which suggested that the slow activation behaviour may be related to the density of the catalyst and the illumination power. The observed activation time was comparable to the timescale we observed in sequential activation (Supplementary Fig. 22 and Supplementary Note 9). However, QCOM offers a spatiotemporal-resolved correlation for this reaction system, complementing the insights into the spatiotemporal dynamics in photocatalysis[31]. Moreover, to gain insights into the nanoscale structure–activity relationship, we imaged these TiO_2 nanowires by electron microscopy methods and observed heterojunctions along a single nanowire, which are known to act as active sites in photocatalysis[37] (Fig. 3i–k and Supplementary Figs. 23 and 24).

In situ spin spectroscopy of free radicals

To further rationalize the reaction mechanism and sequential activation effect, we needed to identify the exact nature of the free radicals (·OH or $O_2^{\cdot-}$) generated at those active sites. However, T1-based imaging reflects the overall presence of free radicals. The NV centre can also be used to probe external electron spins to identify the nature of the free radicals[38], comparable to ensemble ESR spectroscopy but also allowing direct probing at the chemical interface (Supplementary Note 10). However, resolving free-radical electron spins in chemical reactions is challenging because the coherence time of NV centres is limited by additional solution noise interactions under the in situ conditions[39] (Supplementary Figs. 25 and 26 and Supplementary Note 11).

To precisely control the spin states of the NV centre, we performed Rabi and spin-echo measurements to obtain the π/2 time (67 ns; Fig. 4a) and transverse relaxation time (T2, 156.9 μs; Fig. 4b) of the

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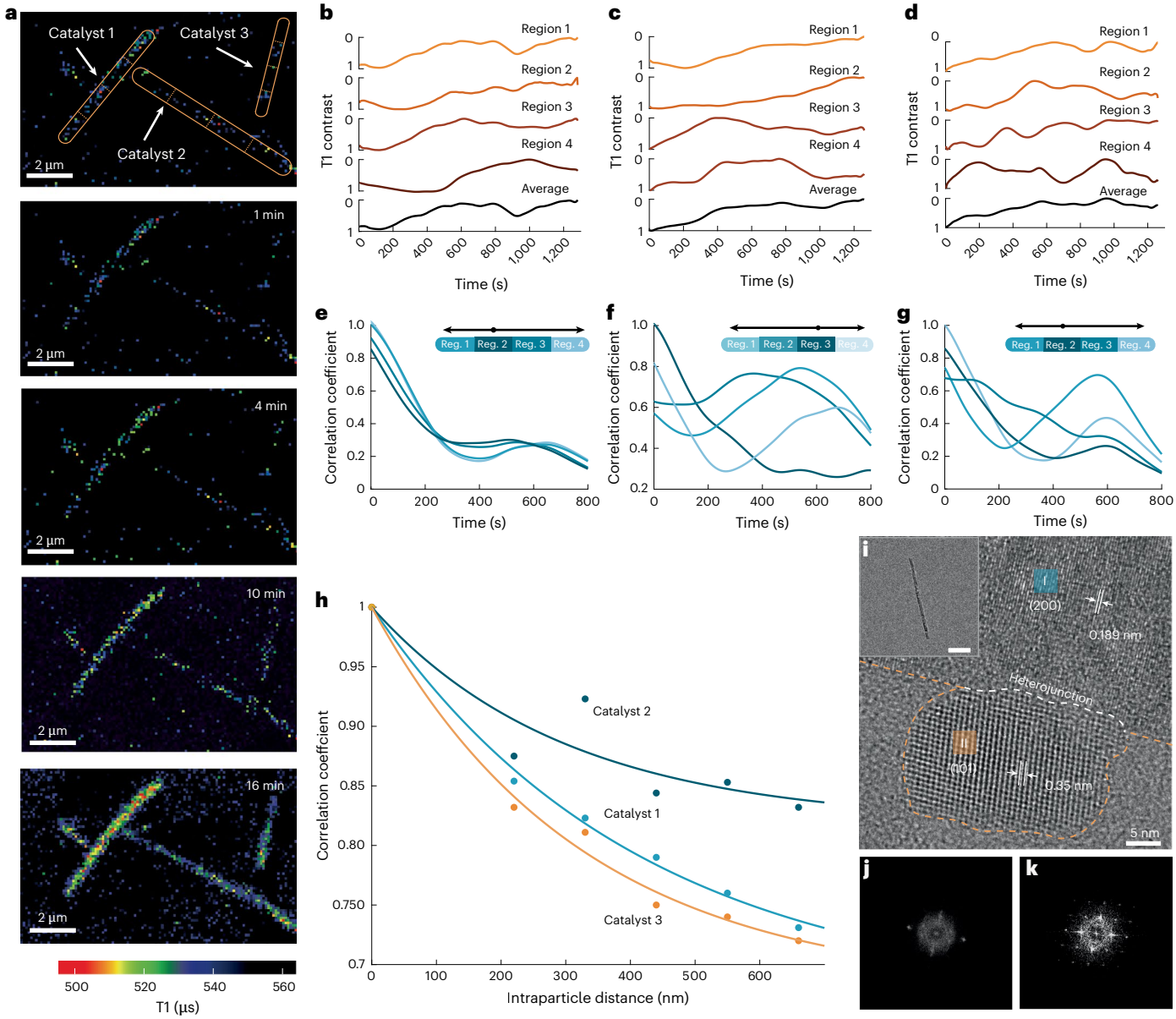

Fig. 3 | Observation of the spatiotemporal sequential activation effect. **a**, Free-radical imaging of single nanowire catalysts at different times (t = 1, 4, 10 and 16 min) during long-term illumination. The top figure schematically illustrate the intraparticle region divisions for the subsequent Pearson cross-correlation analysis. **b**–**d**, Variation in T1 values for four regions of catalysts 1 (**b**), 2 (**c**) and 3 (**d**) in the photocatalytic reaction. Average T1 values for the four regions are also shown for each catalyst. **e**–**g**, Pearson correlation analysis of the curves in **b**–**d** for catalysts 1 (**e**), 2 (**f**) and 3 (**g**), revealing the relationship between the correlation coefficient and time. Reg., region. Dots and arrows represent the starting points and propagation directions of sequential activation effects. **h**, Pearson correlation analysis showing the relationship between the correlation coefficient and intraparticle distance for catalysts 1–3. **i**, Transmission electron microscopy (TEM) image of a heterojunction on a single TiO_2 nanowire. Inset: TEM image of a single TiO_2 nanowire. Scale bar, 200 nm. **j**,**k**, Fast Fourier transform analysis of the regions Ⅰ (**j**) and Ⅱ (**k**) shown in **h** reveals the different crystal faces of these two regions on a single TiO_2 nanowire.

NV centre. We then used in situ double electron–electron resonance (DEER) spectroscopy (Fig. 4c and Methods), where only the target free radicals generated during the chemical reaction can lead to phase accumulation (Fig. 4c). We also recorded the DEER spectrum of the target spin generated in water and trapped by the spin-trapping agent 5,5-dimethyl-1-pyrroline *N*-oxide (DMPO; Fig. 4d). The emerging peaks observed at 105, 148 and 191 MHz (Fig. 4e, obtained under an external magnetic field of 60.2 G), corresponding to the ensemble ESR spectrum in Fig. 4g, match exactly the hyperfine splitting[2,40] between nitrogen and hydrogen in DMPO–OH adducts (Fig. 4f), while the remaining peaks at 130 and 170 MHz originate from the inherent impurity (P1 centre) in diamond[41]. This indicates that it is mainly ·OH that is produced at the active sites of TiO_2 in water.

Owing to its in situ capability, our measurement allows for continuous generation and characterization of free radicals, thereby overcoming the difficulties of detection due to the short lifetime of free radicals[42]. This led to localized label-free ESR spectroscopic analysis of the free-radical species generated at the interface (Fig. 4h,i), which revealed a clear difference between the in situ spectra recorded in water and methanol. We attribute this difference to the fact that methanol, an ·OH sacrificial agent[27], promotes the generation of $O_2^{\cdot-}$ radicals.

Based on the spectral results and imaging data, we can propose a sequential activation mechanism across the active sites: a heterojunction site in a single TiO_2 catalyst is first activated, which then activates a neighbouring site in a sequential manner via hole hopping, reducing water to ·OH (Fig. 4j and Supplementary Fig. 27).

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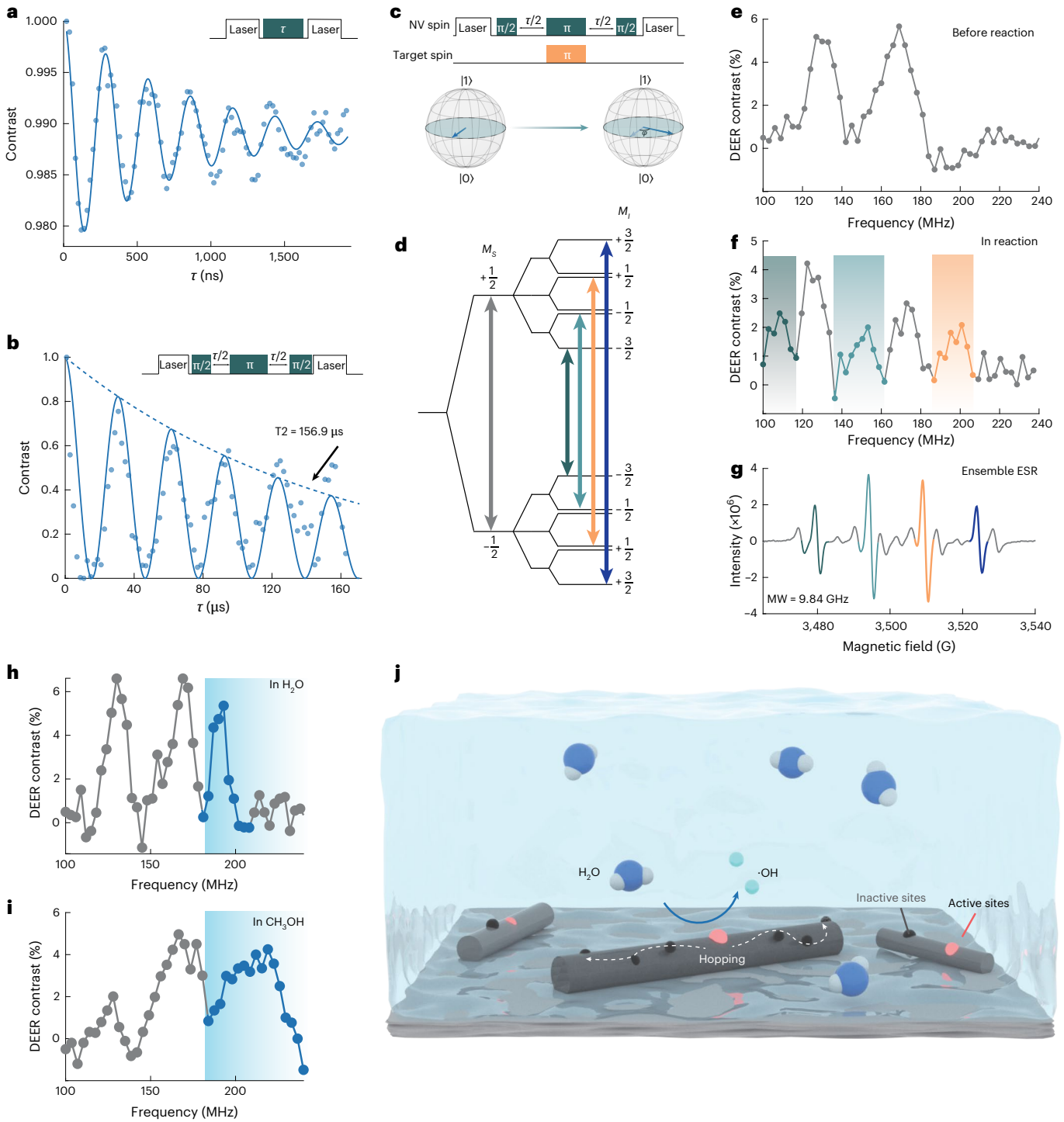

Fig. 4 | Direct operando electron spin spectroscopy of free radicals. a, Rabi measurement of an NV centre using the sequence shown. The data were fitted using a cosine function with exponential damping, from which the π/2 time was fitted as 67 ns. **b**, Spin-echo measurement of an NV centre using the sequence shown. The transverse relaxation time (T2) of the NV centre was fitted as 156.9 μs. **c**, Schematic of the concept of operando DEER spectroscopy, φ refers to the accumulated phase of NV spin under the target spin noise. **d**, Hyperfine splitting energy level diagram of the DMPO–OH adduct. M_S and M_I are the spin and orbital magnetic quantum numbers of the radical electron and nucleus, respectively. **e**,**f**, DEER measurement of an NV centre before (**e**) and during (**f**) the TiO_2 photocatalytic reaction. The shaded bars represent the newly emerging DEER peaks during the reaction process. **g**, Ensemble ESR spectrum of the photocatalytic reaction of TiO_2 in water. **h**,**i**, DEER spectra of the free radicals generated in the chemical reaction performed in H_2O (h) and CH_3OH (**i**) as solvent. The shaded bars represent distinct DEER peaks arising from the reaction processes in two different solvents. **j**, Schematic diagram of the sequential activation mechanism in single TiO_2 nanowires.

Conclusions

Advancing a comprehensive understanding of chemical reactions is never a matter of pursuing single metrics, but rather requires the constraints between multiple observation parameters to be untangled. Building an overview demands measurement capabilities that include spatial imaging, dynamic tracking, the monitoring of multiphysical quantities, operation under diverse conditions and adequate sensitivity, making it a non-trivial task. Notably, our technique allows in situ

imaging with environmental operation (gas, liquid, solid and interface), high spatiotemporal resolution (representative FWHM resolution of ~312 nm and ~10–240 ms), high sensitivity (LOD of around four free radicals) and spectroscopic identification (roughly megahertz spectral resolution). A unique feature of QCOM lies in its multiphysical imaging capability, which can be anticipated to extend its application to image local temperature and electrical and magnetic field changes in a large variety of chemical reaction systems. This provides QCOM with the capacity to observe chemical transformations other than free-radical reactions that are accompanied by local physical field changes. By converging quantum sensing with chemical reaction imaging, our QCOM method offers a powerful in situ method for chemical observation by simultaneously satisfying the criteria of detection sensitivity, resolution and multiphysical imaging, complementing the existing chemical reaction observation toolbox, including single-molecule fluorescence/chemiluminescence imaging[4,10,43], scanning probe microscopy[44,45], in situ electron microscopy[46] and tip-enhanced Raman spectroscopy[47,48] (Supplementary Fig. 28 and Supplementary Table 2). Therefore, we believe that the development of QCOM represents an important step towards unveiling undetected spatiotemporal dynamics in chemistry.

Methods

QCOM set-up

Our home-built QCOM set-up uses a de novo design involving both widefield and confocal modules (see Supplementary Figs. 1 and 2 for details). The microscope is based on a precision displacement control system used for sample positioning and imaging. The system consists of three displacement stages: a piezo stage (P15.XYZ300S-B, Coremorrow) with 8-nm displacement precision in the *x* and *y* directions and 2.5-nm displacement precision in the *z* direction, a displacement stage with 1.5-µm accuracy in the *x* and *y* directions (XWJ-100R-2G, SYMC) and a displacement stage with 1-µm accuracy in the *z* direction (ETSL-20G, SYMC).

A 532-nm continuous-wave laser (MGL-V-532-water-5W, CNI) is passed through an acousto-optical modulator (AUT-AOM 3080-125, G&H) to generate the laser pulse, which is mode-shaped by a fibre (P2-460-HP-3.0-1M-FC/APC, Connect Laser) and a collimator (FP5-F5AP-A, LBTEK). The laser is then reflected by a dichroic mirror (Di02-R532-25×36, Semrock) and directed into a numerical aperture = 1.5 objective (UPLAPO100XOHR, Olympus) to excite a diamond chip for initialization and readout of NV centres. The fluorescence from NV centres is collected by the same objective and filtered by a 549-nm long-pass filter (ff01-549/15, Semrock). It is then split into two parts: one half passes through a pinhole and enters a single photon counter module (SPCM-AQRH-11-FC, Excelitas) whose signal is acquired by a data acquisition card (PCIe-6353, NI), while the other half is received by an sCMOS camera (Prime 95B, Photometrics) for image reconstruction.

MW and radiofrequency (RF) pulses used to drive the NV centres and free radicals, respectively, are generated by two signal generators (N5183B, Keysight) and passed through MW switches (ZYSWA-2-50DR+, Mini-circuits), amplifiers (ZHL-10M4G21W2+ for RF, ZHL16W-43+ for MW, Mini-circuits), combiners (HWDP-DC400-20004600M, Lottak) and circulators (SHX-TG301A, Huaxiang) and then delivered sequentially to a printed circuit board (PCB) and a coplanar waveguide fabricated on a glass substrate. An adjustable 0–250 G external magnetic field is generated by an electromagnet, which aligns the NV *z* axis, as revealed by the ESR spectra. Synchronization is triggered by sending commands to a pulse signal generator (PBESR-PRO-500-USB-RM, PulseBlaster). This ensures the synchronization of all instruments (Supplementary Fig. 2).

Construction of a chemical reaction–quantum sensing interface

The bulk diamond was fabricated by chemical vapour deposition ($3 \times 3 \times 0.1$ mm^3, Ningbo Crysdiam). Ion implantation (ion source: $^{14}N^+$; energy: 10 keV; dose: 1×10^{13} cm^{-2}; angle: 7°; time: 1 h) and subsequent annealing process (850 °C vacuum for 2 h and 500 °C air for 2 h) led to the creation of a shallow layer of NV centres, with the depth below the surface expected to be <25 nm (Supplementary Fig. 4 and Supplementary Note 1). The diamond was first cleaned in piranha solution (98% sulfuric acid–30% hydrogen peroxide, 3:1) at 80 °C for 4 h, then subsequently ultrasonically cleaned with deionized water for 20 min. UV glue was spin-coated at 1,000 r.p.m. onto one side of the cleaned diamond and attached to the inner wall of a self-designed 3D-printed chamber. Then, 20 µl of a 0.1 mg ml^{-1} TiO_2 catalyst solution was applied to the diamond surface and incubated, leaving a layer of catalyst after evaporation of the solvent. Finally, the entire device was glued with polydimethylsiloxane to a coverslip patterned with a fabricated waveguide, and the top of the device was reserved for a piping connection to a peristaltic pump, thus forming an in situ reaction chamber in which the solution can circulate. At the same time, thermocouples and heated ceramic plates were inserted into the solution to monitor and control the temperature. The entire device was connected to the external PCB and fixed to an aluminium tank, which was connected to the displacement stage.

TiO_2 photocatalysis

Two types of catalyst were used in the experiments, that is, anatase-type TiO_2 nanoparticles (XF102, XFNANO) and TiO_2 nanowires (XFJ20, XFNANO). These catalysts were prepared as a 0.1 mg ml^{-1} solution in water before performing experiments. We mounted a 365-nm LED (M365L5, Thorlabs) on the QCOM device as a light source to turn on the photocatalytic reaction, which is controlled by an LED driver (LEDD1B, Thorlabs).

Dynamic imaging of free radicals using the FRI sequence

For the FRI sequence, we first constructed a T1 relaxation sequence based on an sCMOS camera (Fig. 1c). The NV centres were initialized by a series of combinations of laser and dark pulses. To ensure the initialization of the NV centres, the laser pulse length was set to 5 µs. We repeated the sequence and performed 20 measurements by varying the dark pulse time from 2 µs to 2 ms (Supplementary Table 1 and Supplementary Note 5). The time interval was log-linear to coincide with the NV fluorescence decay rate. To ensure an adequate signal-to-noise ratio, we repeated the above measurements 400 times. The T1 relaxation curve was fitted with a biexponential function, and the T1 value was calculated using equations (3) and (4):

$$\mathrm{PL}(\tau) = I_0 + A\mathrm{e}^{-\tau/\mathrm{T1a}} + B\mathrm{e}^{-\tau/\mathrm{T1b}}, \quad (3)$$

$$\mathrm{T1} = \max(\mathrm{T1a}, \mathrm{T1b}), \quad (4)$$

where PL(τ) is the detected fluorescence intensity, I_0 is the baseline value, A and B are prefactors, τ is the length of dark pulse, and T1a and T1b are fitted relaxation times.

Dynamic relaxation imaging was achieved by integrating a 'moving window average' method (Supplementary Figs. 9 and 18 and Supplementary Note 5) and an image denoising technique, which can greatly increase the temporal resolution while maintaining good imaging contrast.

Quantitative modelling of the relationship between T1 and free radicals

The capability of QCOM in free-radical quantification was assessed in three steps. First, the ·OH concentration was calibrated. We used disodium terephthalate as the indicator, which reacts with ·OH in a 1:1 stoichiometric ratio to produce fluorescent HTA (λ_{ex} = 315 nm and λ_{em} = 425 nm). HTA therefore provides a standard curve to quantify the concentration of ·OH. We next measured the fluorescence and NV's T1 value in the same reaction system. The HTA fluorescence intensity

was leveraged to determine the amount of ·OH in the reaction system. Meanwhile, an increasing ·OH concentration would generate a decreasing T1 contrast in QCOM, which was used to build the correlation. Finally, given the correlation data obtained from T1 measurement at different concentrations of ·OH (C_r), to quantitively describe the analytical relationship between them, we developed a theoretical model (see Supplementary Note 6 for details) to estimate the influence of free-radical magnetic noise at NV centres according to equations (2), (5) and (6):

$$B_\perp^2 = \sum_i \left(B_\perp^2\right)^{(i)} = \frac{1}{6}\left(\frac{\mu_0 \gamma_e \hbar}{4}\right)^2 \frac{1}{\pi L^3} C_r, \quad (5)$$

$$\tau_c^{-1} = \frac{\sqrt{2}\mu_0 \gamma_e^2 \hbar}{8\sqrt{\pi}} \frac{C_r^{1/2}}{r_{\min}^{3/2}} + R_{vib}, \quad (6)$$

where $B_\perp^2$ is the variance of the transverse magnetic field at the NV site, τ_c is the correlation time of the free-radical ensemble, $\gamma_e \approx \gamma_{NV} = -2\pi \times 28\ \mathrm{GHz\ T^{-1}}$ is the gyromagnetic ratio, L is the NV depth below the diamond surface, R_{vib} is the vibrational spin relaxation of the radicals, r_{min} is the minimum allowed distance between radicals, μ_0 is the Bohr magneton and $\hbar = h/2\pi$, where h is Planck's constant.

NV-based operando DEER spectroscopy

Experiments were conducted under an external magnetic field of 60 G created by electromagnets, which allowed us to split the degenerate energy levels of NV centres. The NV centres were driven at a frequency corresponding to the resonance frequency determined in the optically detectable magnetic resonance measurements (Supplementary Fig. 25) and we determined the duration of the π-pulse from Rabi oscillation curves (Fig. 4a). The DEER sequence was composed of a spin-echo sequence on an ensemble of NV centres accompanied by a RF pulse (Fig. 4c). We then varied the RF pulse frequencies to conduct the DEER measurements.

Data availability

Raw data comprising a large number of movie files supporting the findings of this study are available from the authors upon reasonable request. Source data are provided with this paper.

Code availability

MATLAB codes for processing the raw data are provided in Supplementary Code 1.

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Author contributions

J.F. conceptualized the project. J.F. and Y.Y. designed the set-up. Y.Y. built the set-up with help from Y.X and S.L. Z. Zhou wrote the sequence and quantitative model. Y.Y., Z. Zhou, F.Z., Z.M., C.Z., W.D., S.L., Y.Z., Z. Zhang and W.Z. conducted and analysed the experiments. J.F. supervised the work. J.F. and Y.Y. wrote the paper with feedback from other authors.

Competing interests

J.F. and Y.Y. are listed as inventors in a filed patent (Publication number: CN120102536A) describing the quantum-enabled in-situ imaging technique. The other authors declare no competing interests.

Additional information

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